

PulseEDU

Workflow Guide

Screen-by-screen, function-by-function, per-role reference for the engineering team.

This document maps every screen in the PulseEDU staff and parent applications to the functions exposed on it, and for each function describes what each user role can do. It is intended as a baseline test plan for QA and as a single-source-of-truth for product handoffs to new engineers.

Generated: 2026-05-18

Source: lib/db/src/schema, artifacts/api-server/src/routes, artifacts/client/src/components

How to read this guide

The guide is organized top-down: Authentication, the App Shell, then each major product area as its own section. Inside each section you will find one or more screens. Inside each screen:

- Purpose — what the screen is for, in one paragraph.
- Visibility — which roles can navigate to it at all.
- Functions — every action a user can take on the screen, with a per-role behavior list.
- Notes — gotchas, dependencies, and known follow-ups.

Every per-role line answers two questions: 'is this role even allowed to see this affordance?' and 'if yes, what is the exact behavior they get?'. When a row says a role is hidden, the affordance must not render — and the matching server route must return 403.

QA pass: walk every screen with each role profile in Settings !' Staff Preview. The visible affordances must match the rows in this document.

User Roles Glossary

Roles are additive: SuperUser implies Admin everywhere; District Admin implies Admin within the district. Several capability flags (cap_manage_displays, etc.) can grant a teacher narrow extra access without elevating them to a full role.

SuperUser

Scope: District-wide (all schools, all districts)

Engineering / district-IT level. Sees Tenancy panel, can switch between any school, manages district integrations, can grant/revoke any role. Implicitly counts as Admin everywhere.

District Admin

Scope: District-wide (all schools in their district)

Cross-school analytics and policy. District Overview view aggregates every school side-by-side; can read/write everything in their district but does not manage tenancy.

Admin

Scope: Single school

Principal / AP. Full edit on every tile in their school: staff roles, settings, signage, store, safety plan library, case investigations (including admin-only video evidence + AI Consistency Check), reopening cases.

Dean

Scope: Single school

Investigator without full admin authority. Sees the Investigations Hub, can collect statements, manage cases, run investigations and footage requests. Cannot edit catalog/settings tiles unless also granted Admin.

Behavior Specialist

Scope: Single school

Core Team member. Reads/writes every teacher's Tier 2/3 entries, manages strategy catalog, sees Behavior Specialist hub, MTSS plans, investigations (as a Case Investigator). Manages bell schedules and displays.

MTSS Coordinator

Scope: Single school

Core Team. Same intervention/case investigator privileges as Behavior Specialist; owns MTSS Plan creation/editing and the MTSS Coordinator hub.

School Psychologist

Scope: Single school

Core Team for intervention purposes (reads/writes any teacher's tier work). Excluded from the discipline-investigation chain — does not see the Case Investigator privileges of Dean/BS/MTSS for investigations.

Guidance Counselor

Scope: Single school

Owns Safety Plans (alongside Core Team). Sees Safety Plan editor + library; not an automatic Core Team member otherwise.

PBIS Coordinator

Scope: Single school

Owns the PBIS Hub: thresholds, school-wide expectations, school store catalog, PBIS Points dashboards, and house standings on signage.

Teacher

Scope: Their classes / their roster

Default staff role. Logs hall passes, tardies, PBIS points, accommodation events, interventions for their own students. Read-only on School Store and most catalogs. Sees their Teacher Roster.

Non-Exempt

Scope: Single school

FLSA non-exempt staff whose entire app surface is timekeeping. Sidebar collapses to Hall Pass + Tardy Pass + Comp Time ONLY (including Quick Access — Teacher Roster, PBIS, Request Pullout, etc. all hidden). Applying the preset auto-flips `exempt_status='non_exempt'` so Comp Time accrues. Admin tier escapes the sidebar collapse if accidentally marked Non-Exempt. `exempt_status` is also an independent admin toggle for staff who are non-exempt but don't take this role bundle (e.g., a non-exempt aide who is also a Behavior Specialist).

Front Office

Scope: Single school

Clerical / receptionist staff. Sees everything a Teacher sees (Hall Pass, Tardy Pass, Family Communication, PBIS Points, School Store, Accommodations, Log/My Interventions, AST, Comp Time) EXCEPT Request Pullout — pullouts are a teacher referral, not a front-desk action. Watchlists and Accommodations come through the teacher baseline. Does NOT grant AST or Comp Time approval rights; Confidential Secretary keeps its existing `canApproveAst` grant unchanged.

SRO

Scope: Single school

School Resource Officer (sworn officer assigned to the school). Same capability bundle as Teacher — action-capable on Hall Pass, Tardy Pass, PBIS award, Family Communication, etc. Broken out as its own role so future SRO-specific surfaces (incident logs, weapon screenings) can target it cleanly.

Guardian

Scope: Single school

Hall monitor / security aide / campus guardian. Same capability bundle as Teacher today. Broken out as a distinct role for reporting and future role-targeted features.

Parent

Scope: Their student(s) only

Lives in `/parent/*` portal — separate from the staff app. Magic-link / invite-based login. Reads HeartBEAT data (PBIS, hall passes, tardies, accommodations, staff notes) for any student linked to their account; can switch between siblings.

1. Authentication & First Run

Every staff role uses the same sign-in screen; routing/visibility branches once the session is established. Parents sign in through a separate /parent/* path with magic-link invites managed by Admins.

Staff Sign-in

Nav path: / (root, when unauthenticated)

Audience: All staff roles

Purpose

Authenticate a staff member into a single school context. Multi-school users (SuperUser, District Admin) land on a school switcher after auth.

Function — Sign in

Email + password (or SSO if district has it configured).

Per-role behavior:

All staff — Authenticate; failed attempts are rate-limited per server policy.

Function — Pick active school

After login, multi-school accounts pick which school context to operate in. Single-school accounts skip this.

Per-role behavior:

SuperUser — See every school across every district.

District Admin — See every school in their assigned district.

Admin, Dean, Behavior Specialist, MTSS Coordinator, Teacher — Skip — auto-bound to their assigned school.

Notes:

- School context lives in middleware as req.schoolId; every server query filters by it.

Onboarding Checklist

Nav path: Auto-displays on first login for Admins until cleared

Audience: Admins only

Purpose

Phase-by-phase setup wizard so a new school comes online without missing a configuration step (bell schedule, locations, kiosks, signage, PBIS thresholds, etc.).

Function — Tick a step complete

Marks a row as done; persists on the school_onboarding_state table.

Per-role behavior:

Admin — Tick any step; only Admins see the checklist.

Teacher, Dean, Behavior Specialist, MTSS Coordinator, Guidance Counselor — Do not see the checklist at all.

Function — Export onboarding PDF

Generates a printable copy of the checklist with 'how this works' copy alongside each step. Useful for handoff to a co-administrator.

Per-role behavior:

Admin — Download the PDF (server-rendered via pdfkit).

2. App Shell — Sidebar, Quick Access, Top Bar

The sidebar is feature-flag aware and role aware. Every nav item is hidden if the user lacks the relevant capability; the visible set is the user's effective toolbox. School-level feature flags can hide entire modules even from Admins.

Left Sidebar

Nav path: Always visible (collapsible)

Audience: All staff

Purpose

Primary navigation. Items are grouped: Daily Ops (Hall Passes, Tardies, PBIS, Family Comm), Interventions, Investigations, Insights, Admin. Each group is hidden if the user has no items in it.

Function — Quick Access (top of sidebar)

Promotes 1–3 high-priority items based on pending work. Verify Pullouts gets promoted here when the queue is non-empty (replaces its later in-list slot to avoid duplication).

Per-role behavior:

Teacher — Sees Hall Passes, Tardies, PBIS Points, My Interventions, Family Comm (per feature flags).

Dean — Sees Investigations, Hall Passes, Verify Pullouts (when pending).

Admin — Sees everything that is enabled for the school plus admin-only entries.

Function — Investigations entry

Routes to Watchlist Hub. Visibility-gated.

Per-role behavior:

Admin, SuperUser, District Admin, Dean, Behavior Specialist, MTSS Coordinator — See the entry.

Teacher, Guidance Counselor, PBIS Coordinator — Do not see Investigations in the sidebar.

Function — Settings entry

Routes to Settings Hub (tile launcher).

Per-role behavior:

Admin, SuperUser, District Admin — Always visible.

Behavior Specialist, MTSS Coordinator — Visible — limited tile set (bell schedules, intervention strategies, etc.).

Teacher — Hidden.

Screen notes:

- Nav-key feature map: hallPasses !' HallPasses, pbis !' Pbis, schoolStore !' SchoolStore, etc. Admins can disable a feature for their school in Settings !' Feature Configuration; the matching nav item disappears for everyone, including Admins, until re-enabled.

School Switcher (top bar)

Nav path: Top bar, only for multi-school accounts

Audience: SuperUser, District Admin

Purpose

Re-bind the active school context without re-authenticating. Updates req.schoolId for every subsequent request.

Function — Switch school

Persists the chosen school in the staff session. All open tabs/sections re-fetch with the new schoolId.

Per-role behavior:

SuperUser — Switch to any school in any district.

District Admin — Switch to any school in their district.

Admin, Teacher, Dean, Behavior Specialist, MTSS Coordinator — No switcher — single-school binding.

3. Daily Operations

The screens teachers and front-office staff use every period.

Hall Passes

Nav path: Sidebar !' Hall Passes

Audience: All staff (teachers most often)

Purpose

Issue a hall pass to a student, watch the clock, and check them back in. Powers the in-room flow plus the kiosk signage.

Function — Create pass

Pick student, destination, optional note. Creates a hall_passes row, may notify the destination teacher's allowlist.

Per-role behavior:

Teacher — Create a pass for any student in their roster; destination must be on their teacher allowlist if configured.

Admin, Dean — Create passes school-wide (any student, any destination), bypassing teacher allowlist.

Function — Check in / out

Stamps left_at and arrived_at, optionally returned_at. Drives stale/red highlighting after threshold.

Per-role behavior:

Teacher — Check in/out passes they own or that arrive at their room.

Admin, Dean — Override any pass.

Function — Hall Pass Queue

School-wide queue of active passes. Auto-resets per bell-schedule period (or 45-min idle bucket as fallback). Used at the front desk to spot-check who is out of class.

Per-role behavior:

Admin, Dean, Behavior Specialist, MTSS Coordinator — View and manage.

Teacher — View only.

Notes:

- Requires a default bell schedule for period-based reset; otherwise falls back to idle-time bucketing.

Function — Per-student pass limit

Caps how many passes a student can hold open / per day.

Per-role behavior:

Admin — Configure global default in Settings !' School.

Teacher — Override per student via Roster row.

Tardy Pass

Nav path: Sidebar !' Tardy Pass

Audience: All staff

Purpose

Quick capture of late arrivals tied to a period from the bell schedule.

Function — Log tardy

Pick student + period; writes a tardies row. Aggregated into Insights !' Engagement and the Parent Portal.

Per-role behavior:

Teacher — Log tardies for their own students.

Admin, Dean — Log for any student.

Function — Edit / delete

Correct a wrong-student or wrong-period entry.

Per-role behavior:

Teacher — Edit their own entries (audit-logged).

Admin — Edit/delete any entry.

PBIS Points

Nav path: Sidebar !' PBIS Points

Audience: All staff

Purpose

Award positive-behavior points to students using school-defined reasons. Drives house standings on signage and the school/classroom store ledger.

Function — Award points

Pick student(s) + reason + optional note. Multi-select supported for whole-class awards.

Per-role behavior:

Teacher — Award to any student in their roster.

Admin, Dean, PBIS Coordinator — Award school-wide.

Function — Note templates

Reusable pre-written notes for common awards (e.g. 'On-task during fire drill'). Speeds up entry on slow devices.

Per-role behavior:

PBIS Coordinator — Manage templates.

Teacher — Pick from templates.

Function — Reasons catalog

School-wide list of award reasons (e.g. 'Showed leadership').

Per-role behavior:

PBIS Coordinator — Add / edit / retire reasons.

Teacher — Use only.

School Store (read-only catalog)

Nav path: Sidebar !' School Store

Audience: All staff

Purpose

Browse the school-wide rewards catalog so teachers can answer 'what can I redeem?' from the classroom.

Function — Browse items

Lists school store items with cost, image, stock.

Per-role behavior:

All staff — Read-only browse.

Function — Redeem

Spend a student's PBIS balance for an item; deducts points and writes a store_orders row.

Per-role behavior:

PBIS Coordinator, Admin — Redeem on a student's behalf.

Teacher — View only — redemption happens at the PBIS Hub or at admin.

Classroom Store

Nav path: Inside PBIS Hub / per-teacher

Audience: Teachers

Purpose

Per-teacher reward catalog (unique to each teacher). Lets teachers run a parallel economy alongside the school store.

Function — Manage items

CRUD on classroom store items with thumbnail, cost, stock.

Per-role behavior:

Teacher — Full edit on their own classroom store.

Admin — Read across all teacher catalogs (audit).

Function — Redeem

Spend points for a classroom item.

Per-role behavior:

Teacher — Redeem for their own students.

Spotlight

Nav path: Header pill / Quick Access

Audience: All staff

Purpose

Recognize a student publicly — short message + optional photo that surfaces on signage.

Function — Launch spotlight

Pick student, write message, optional emoji. Appears in signage rotation immediately.

Per-role behavior:

Teacher — Spotlight any student in their roster.

Admin, PBIS Coordinator — Spotlight any student in the school.

Accommodations

Nav path: Sidebar !' Accommodations

Audience: All staff who teach the student

Purpose

Log when an accommodation was provided (extended time, fidget, separate room, etc.) for IEP/504 evidence.

Function — Log accommodation use

Pick accommodation + student. Stamps the date and optional context.

Per-role behavior:

Teacher — Log for their own students.

MTSS Coordinator, Admin — Log school-wide.

Function — View accommodation log

Chronological list per student.

Per-role behavior:

All staff who can see the student — Read.

Parent — Read for their child via Parent Portal (if enabled in Parent Portal Sections).

4. Interventions (Tier 2 / Tier 3 / MTSS)

The intervention stack: who logs it, who tracks completion, who reads the strategy catalog.

Log Intervention

Nav path: Sidebar !' Log Intervention (tier-aware launcher)

Audience: All staff

Purpose

Single launcher that figures out whether the picked student is on a Tier 2 or Tier 3 plan and routes the form accordingly.

Function — Pick student

Searches the student finder; surfaces tier badge.

Per-role behavior:

Teacher — Search across their roster.

Core Team — Search across the entire school.

Function — Submit Tier 2 daily entry

Daily check-in / check-out signal (CICO) form: morning rating, afternoon rating, behavior notes.

Per-role behavior:

Teacher — Submit for their own students.

Core Team — Submit on behalf of any teacher.

Function — Submit Tier 3 weekly progress

Weekly ratings against goal categories with strategy-used checkboxes.

Per-role behavior:

Teacher — Submit for their own students.

Core Team — Submit for any student.

My Interventions

Nav path: Sidebar !' My Interventions

Audience: All staff

Purpose

Personal queue: which of MY students are owed an intervention check-in today? Surfaces the count in the Bell notification.

Function — Today list

Filterable list of owed entries grouped by student.

Per-role behavior:

Teacher — See their own owed list.

Core Team — See their own AND a 'Today across school' aggregate.

Function — Bell notification

Sidebar bell shows count of owed entries; clicking jumps here.

Per-role behavior:

All staff with owed entries — See the bell.

Intervention Reports

Nav path: Sidebar !' Interventions (Core Team only)

Audience: Core Team

Purpose

Completion-rate report and tier roll-ups across all teachers.

Function — Completion report

% of expected entries actually logged per teacher / per student / per week. Drill-in to see missed days.

Per-role behavior:

[All Core Team](#) — View.

[Teacher](#) — Hidden.

Function — Strategy catalog

School-wide list of Tier 3 strategies grouped by category (e.g. self-regulation, social skills).

Per-role behavior:

[All Core Team](#) — Add / edit / retire categories and strategies.

[Teacher](#) — View only via the Tier 3 weekly form.

MTSS Plans

Nav path: Sidebar ! MTSS Coordinator hub

Audience: MTSS Coordinator + Admins

Purpose

Tier 2 / Tier 3 intervention plan tracking: goal setting, weekly progress monitoring, completion reports.

Function — Create plan

Pick student, set tier, choose goal categories, assign owner.

Per-role behavior:

[MTSS Coordinator](#) — Create / edit / archive any plan.

[Admin](#) — Same.

[Teacher](#) — View their own students' active plans (read-only).

Function — Goal version history

Tier 3 goals are append-only versioned — editing a goal creates a new version with effective_from timestamp.

Per-role behavior:

[Core Team](#) — Edit (creates new version).

Function — Weekly progress monitor

Rating + note per goal per week.

Per-role behavior:

[Teacher](#) — Submit for their own students on the plan.

[Core Team](#) — Submit on behalf of teacher.

Request Pullout

Nav path: Sidebar ! Request Pullout

Audience: Teachers

Purpose

Teacher requests an intervention pullout for a specific student/period. Admins/Core Team triage and verify in the Verify Pullouts queue.

Function — Submit request

Pick student + reason from the pullout reasons catalog + period.

Per-role behavior:

[Teacher](#) — Submit.

Function — Verify Pullouts (admin queue)

Queue view of pending requests with approve/reject. Promoted to Quick Access in the sidebar when count > 0.

Per-role behavior:

Admin, Behavior Specialist, MTSS Coordinator — Verify, approve, reject, add note.

Teacher — See status of their own requests only.

5. Safety Plans

Per-student behavioral / physical safety checklists with a school-wide library.

Safety Plan Editor

Nav path: Click red 'SP' pill on the Teacher Roster, or via Investigations

Audience: Guidance Counselor + Core Team

Purpose

Build/edit a student's active safety plan from the school-wide item library.

Function — Add item

Pick from the school's library; or write a free-text custom item. Each item carries a category (e.g. de-escalation, restraint protocol).

Per-role behavior:

Guidance Counselor, All Core Team — Add/edit/remove items, mark plan active/inactive.

Teacher — View the active plan for their own students (read-only).

Admin, Dean — Read across school.

Function — Audit log

Every change is timestamped with author. Append-only.

Per-role behavior:

All who can read the plan — View history.

Function — Library management

School-wide reusable items.

Per-role behavior:

Guidance Counselor, All Core Team — CRUD on the library.

Teacher — Hidden.

Screen notes:

- Active safety plan presence drives the red 'SP' indicator on the Teacher Roster and Student Profile so teachers know to review before a difficult interaction.

6. Investigations (Watchlist Suite)

The discipline investigation stack — Hub, Schoolwide Behavior Network, Case Detail (with Investigation tab), Student Graph. Visibility is gated on Admin / Dean / Behavior Specialist / MTSS Coordinator (the 'Case Investigator' set), with admin-only sub-affordances inside.

Investigations Hub (Watchlist Hub)

Nav path: Sidebar !' Investigations

Audience: Case Investigators (Admin / Dean / Behavior Specialist / MTSS Coordinator) + SuperUser / District Admin

Purpose

Mission control for behavior cases — alerts, orbit (students in flight), active cases, recent incidents, witness statements queue.

Function — Active cases panel

Top-6 active cases (open/monitoring/escalated) by default. Inline search box + status pill row (Active / Open / Monitoring / Escalated / Closed / All) reveals filtered results live.

Per-role behavior:

Investigators — Open any case from the panel.

Admin — Plus the '+ New' button to launch the New Case modal.

Notes:

- Search matches case title, case number (full '25-26-0042' or bare '42'), and lead staff name.
- Slice-to-6 only applies when the default Active filter is selected and search is empty; otherwise all matches show.

Function — Orbit list

Students currently 'in flight' in any active case. Sphere icon next to each links to their Student Graph.

Per-role behavior:

Investigators — Click to drill in.

Function — Statements intake

Pending witness statements with status (requested / reminded / completed / waived) + Dismissed tab for triaged-out statements that can be restored.

Per-role behavior:

Investigators — Send reminders, mark complete, dismiss with reason.

Admin — Plus restore-from-dismissed.

Function — Recent incidents feed

Loose interactions not yet promoted to a case. Includes severity color coding.

Per-role behavior:

Investigators — Promote to a new case (opens Promote-to-Case modal) or attach to an existing case.

Admin — Same plus delete.

Function — Open Schoolwide Behavior Network button

Routes to the network visualization.

Per-role behavior:

Investigators — Click.

Schoolwide Behavior Network

Nav path: Investigations Hub !' 'Open Network' button

Audience: Case Investigators

Purpose

School-wide spider-web map of who keeps showing up together in incidents. Each active case is its own ring of student spheres; the loose ring at the top holds students with interactions but no case.

Function — Time window filter

Restrict to last N days (default 30).

Per-role behavior:

Investigators — Change window.

Function — Click sphere

Opens the Student Graph for that student.

Per-role behavior:

Investigators — Drill in.

Function — Click case label

Opens the Case Detail.

Per-role behavior:

Investigators — Drill in.

Function — Sphere sizing

Adapts to spiral spacing (clamp 0.45–1.6 of base) so high-frequency students visibly pop on dense schools without overwhelming a sparse view.

Per-role behavior:

All viewers — Read sizing as visual cue.

Case Detail

Nav path: Hub !' click case row, or Network !' click case label

Audience: Case Investigators (visibility) + Admins (admin-only sub-affordances)

Purpose

All artifacts of one investigation: incidents, participants, statements, video evidence, AI consistency check, notes, audit log.

Function — Tab nav: Overview / Investigation

Overview is the default participant + incidents list. Investigation tab launches the per-incident ring (see next screen).

Per-role behavior:

All who can see the case — Switch tabs.

Function — Edit title / summary / lead

Inline edit with audit row.

Per-role behavior:

Admin, Dean, Behavior Specialist, MTSS Coordinator — Edit.

Function — Change status

Move between open / monitoring / escalated. Closing requires the dedicated Close modal (cannot be done via this dropdown).

Per-role behavior:

Investigators — Move between non-closed statuses.

All — Cannot reopen via PATCH — server rejects status='open' on a currently-closed case and directs to /reopen.

Function — Close case (modal)

Opens CloseCaseModal: required outcome dropdown (from the school's catalog), required note when outcome is 'other'. Writes outcome_code, outcome_note, closed_by_*, audit row.

Per-role behavior:

Investigators — Close.

All — Cannot close without picking an outcome — server enforces.

Function — Reopen case

Admin-only. Prompts for a reason ("e5 chars). Preserves prior outcome metadata; flips status to open and writes audit row.

Per-role behavior:

Admin, SuperUser, District Admin — Reopen.

Dean, Behavior Specialist, MTSS Coordinator — Hidden — server returns 403.

Function — Witness statements panel

Per-incident statements with reader, complete/remind/dismiss/restore.

Per-role behavior:

Investigators — Manage.

Function — @-mention chips in statements / notes

Typing @ inserts a structured chip (Display Name | STUDENTID). Persisted in the body and indexed in case_mentions.

Per-role behavior:

Anyone editing statement / note — Insert chips.

Function — MentionSuggestStrip

After typing/save preview, AI scans the body against the roster and proposes additional mentions ('We think this also references X, Y'). One-click insert.

Per-role behavior:

Anyone editing — See suggestions; suggestions are NOT auto-confirmed — they only become structured mentions when the user clicks insert.

Function — AI Consistency Check (admin-only)

Header pill + side panel + per-row dot. Runs Anthropic against statements within the case, flags contradictions/corroborations.

Per-role behavior:

Admin, SuperUser, District Admin — Run / view findings; dismiss with justification.

Dean, Behavior Specialist, MTSS Coordinator — Hidden.

Notes:

- Phase-3 of the case suite. Dismissal is persistent suppression scoped to the case.

Function — Video Evidence panel (admin-only)

Upload / link camera feed clips, tag students appearing in the clip.

Per-role behavior:

Admin, SuperUser, District Admin — Manage.

Other Investigators — Hidden.

Function — Case notes (timeline)

Append-only running notes (meeting summaries, parent calls, room moves).

Per-role behavior:

All who can see the case — Add notes.

Function — Footage requests panel

Request that the camera operator pull a window of footage; tracks status.

Per-role behavior:

Investigators — Submit / view.

Investigation Tab (per-incident ring)

Nav path: Case Detail !' Investigation tab

Audience: Case Investigators

Purpose

Per-incident witness graph. Center = the incident; rings = principals, witnesses with statements, mentioned-but-silent students.

Function — Incident selector

Pick which incident on the case to render.

Per-role behavior:

Investigators — Switch incidents.

Function — Ring layout

Anchor (incident) at center; principal participants closest; witness-statement authors next; mentioned-but-silent on the outer ring.

Per-role behavior:

Investigators — Read.

Function — Edges

Drawn ONLY from confirmed structured @-mentions in case_mentions, plus corroborate/contradict pairs from case_consistency_findings. AI suggestions never become edges until confirmed.

Per-role behavior:

All viewers — Read edges as evidence trail.

Function — Side panel (statement reader)

Reuses the case detail's statement reader; shows the highlighted mention's neighborhood.

Per-role behavior:

All who can see the statement — Read.

Function — Request statement (mentioned-only spheres)

If a student is mentioned in someone else's statement but has no statement of their own, a 'Request statement' button appears on their sphere.

Per-role behavior:

Investigators — Send request — creates a witness_statements row in 'requested' state.

Student Graph

Nav path: Network sphere click, or Hub orbit click

Audience: Investigators

Purpose

Single-student spider — every case the student touches and the other students they overlap with.

Function — Open case from graph

Click any case label.

Per-role behavior:

Investigators — Drill in.

7. Insights Dashboards

Aggregate analytics across the school. All dashboards share grade and time-window filters and support drill-down to a Student Profile.

Insights !' Student Watchlist

Nav path: Sidebar !' Insights !' Watchlist

Audience: All staff

Purpose

Per-staffer watchlist of students they want to keep an eye on; cross-references HeartBEAT data.

Function — Add student to my watchlist

Star a student to subscribe to their daily roll-up.

Per-role behavior:

All staff — Manage their own list.

Function — Group entries

Bucket your watchlisted students by tag (e.g. 'AM Tier 2 cohort').

Per-role behavior:

All staff — Manage their own groups.

Engagement Dashboard

Nav path: Sidebar !' Insights !' Engagement

Audience: All staff

Purpose

Tardies, attendance, hall-pass time-out-of-class. Top-N lists per grade with disaggregation by demographic (when configured).

Function — Filters

Grade band, time window, optional demographic disaggregation.

Per-role behavior:

All staff — Filter.

Function — Drill-in to student

Click a row to open Student Profile with the same time window pre-applied.

Per-role behavior:

All staff — Drill.

Behavior Dashboard

Nav path: Sidebar !' Insights !' Behavior

Audience: All staff

Purpose

Discipline incidents, escalations, repeat actors, location heat.

Function — Top-N

Repeat actors / locations.

Per-role behavior:

All staff — View.

Function — Drill-in

Open the case file or student profile.

Per-role behavior:

[Investigators](#) — Open case file.

[Teacher](#) — Open student profile only.

Academics Dashboard

Nav path: Sidebar !' Insights !' Academics

Audience: All staff

Purpose

FAST scores and academic trajectory.

Function — Trajectory chart

Per-window FAST scores plotted with grade-level reference lines.

Per-role behavior:

[All staff](#) — View.

SEB / SEL Dashboard

Nav path: Sidebar !' Insights !' SEB/SEL

Audience: Core Team + Counselors

Purpose

Social-emotional and behavioral screener data, intervention overlay.

Function — Cohort view

Filter to students with screener scores in a band.

Per-role behavior:

[Core Team + Guidance Counselor](#) — View.

Equity Dashboard

Nav path: Sidebar !' Insights !' Equity

Audience: Admins + District Admin

Purpose

Disparities across race / IEP / 504 / ELL bands for discipline and engagement.

Function — Disaggregation table

Side-by-side rates by demographic.

Per-role behavior:

[Admin](#), [District Admin](#), [SuperUser](#) — View.

[Teacher](#), [Dean](#) — Hidden by school policy gate (configurable).

Early Warning Dashboard

Nav path: Sidebar !' Insights !' Early Warning

Audience: Core Team

Purpose

Composite indicators (attendance, behavior, academic, intervention completion) flagging students sliding off track.

Function — Threshold tuning

Score weighting per indicator. Persisted per-school in Settings !' PBIS Thresholds and equivalent.

Per-role behavior:

[Admin](#) — Tune.

[Core Team](#) — View.

Student Profile

Nav path: Drill-in from any dashboard or roster

Audience: All staff who can see the student

Purpose

Single-student 360°: HeartBEAT, FAST, accommodations, safety plan badge, current cases, intervention plans, parent contacts.

Function — Tabs

Engagement / Behavior / Academics / SEB / Interventions / Notes.

Per-role behavior:

[All staff who teach or work with the student](#) — Read.

Function — Add staff note

Free-text note pinned to the student's profile.

Per-role behavior:

[Teacher](#) — Add notes for their own students.

[Core Team](#) — Add for any student.

Function — Open case from profile

Direct link to any case the student is on.

Per-role behavior:

[Investigators](#) — Open.

8. Teacher Roster

The day-to-day teacher view of their students.

Teacher Roster

Nav path: Sidebar !' Quick Access !' Roster

Audience: Teachers (their own); Core Team (any teacher's)

Purpose

List view of students the teacher is responsible for, with FAST scores, ESE/504/ELL flags, safety plan indicator, and quick actions.

Function — Row actions

Pass / Tardy / PBIS / Spotlight / Open Profile / Open Safety Plan (when present).

Per-role behavior:

Teacher — Use any action on a student in their own roster.

Core Team — Same on any teacher's roster.

Function — FAST scores column

Most recent score with delta from prior window.

Per-role behavior:

All viewers — Read.

Function — Program flags

ESE / 504 / ELL pills next to the name.

Per-role behavior:

All viewers — Read.

Function — Safety Plan pill ('SP' red badge)

Indicates the student has an active safety plan; click to open editor.

Per-role behavior:

Teacher — Click !' opens read-only viewer.

Guidance Counselor, Core Team — Click !' opens editor.

Function — Switch to another teacher's roster

Dropdown at top.

Per-role behavior:

Teacher — Hidden.

Core Team — Switch to any teacher in the school.

9. Admin Hub & ISS

Admin-facing operational dashboards: ISS log, recent admin actions, Verify Pullouts, ISS Dashboard.

Admin Hub

Nav path: Sidebar !' Admin Hub

Audience: Admin (and SuperUser/District Admin)

Purpose

Recent admin activity feed: ISS assignments, OSS, dismissals, role changes.

Function — Recent feed

Reverse-chron list of admin actions. Click row !' opens the underlying record.

Per-role behavior:

[Admin](#), [SuperUser](#), [District Admin](#) — View any action.

Function — ISS log row !' detail (planned)

Future: click an ISS assignment to see the full record with edit/delete affordances guarded by 'has any day been served yet'. Required reason-for-edit prompt; audit row written. (See replit.md !' Future work.)

Per-role behavior:

[Admin](#) — Will edit when shipped.

ISS Dashboard

Nav path: Sidebar !' ISS Dashboard

Audience: Admin + ISS room staff (capability-flag granted)

Purpose

Daily roster of students serving in-school suspension; per-period attendance marking, served/rolled state, redirect to OSS handoff if needed.

Function — Mark period present

Updates iss_attendance_day row.

Per-role behavior:

[ISS room staff](#) — Mark present per period.

[Admin](#) — Mark + override.

Function — Roll forward

Carry uncompleted day to next school day.

Per-role behavior:

[Admin](#) — Roll.

Verify Pullouts

Nav path: Sidebar !' Quick Access (when pending) or Verify Pullouts entry

Audience: Admin / Behavior Specialist / MTSS Coordinator

Purpose

Approve teacher-submitted intervention pullout requests.

Function — Approve / Reject

Final disposition + optional note. Approved pullouts surface in My Interventions and student plan.

Per-role behavior:

[Admin](#), [Behavior Specialist](#), [MTSS Coordinator](#) — Approve / reject.

10. Displays / Signage

TV signage in hallways and the kiosk experience.

Displays (admin)

Nav path: Sidebar !' Displays

Audience: Admin / Core Team / staff with cap_manage_displays

Purpose

Per-school playlists for digital signage TVs (image, video, audio, PDF).

Function — Manage playlist

Add / reorder / schedule items.

Per-role behavior:

Core Team — Manage.

Teacher with cap_manage_displays — Manage.

Function — Per-display override

Pin a single item or freeze the rotation on a specific TV.

Per-role behavior:

Admin — Override.

Function — Heartbeat sections

Curated cards (PBIS standings, hall pass live count, etc.) configured per school.

Per-role behavior:

Admin — Configure visibility.

Display Show (TV-facing)

Nav path: /signage/* (no auth — opens via signed kiosk URL)

Audience: Public TV viewers

Purpose

Read-only fullscreen rotation of the playlist + Heartbeat cards.

Function — Auto-rotate

Cycles items per schedule.

Per-role behavior:

Public viewers — Watch.

Function — Active hall pass strip

Live count of out-of-class students.

Per-role behavior:

Public viewers — Watch.

Kiosk Banner / Activation

Nav path: Settings !' Kiosk Setup

Audience: Admin

Purpose

Provisioning a hallway tablet/computer as a hall pass kiosk with a stable token.

Function — Activate kiosk

Generates a kiosk activation token; the device exchanges it for a long-lived viewer token.

Per-role behavior:
Admin — Activate.

11. Data Importer

CSV upload pipeline for assessments, rosters, and behavior data.

Data Imports

Nav path: Settings !' Data Imports

Audience: Admin

Purpose

Upload CSV !' pick template !' preview mapped rows !' commit. Each import is rollback-able while still in the audit window.

Function — Upload CSV

Drop file, choose template (Assessments / Roster / Behavior).

Per-role behavior:

[Admin](#) — Upload.

Function — Preview & map

Show first N rows mapped to columns; warn on type mismatches.

Per-role behavior:

[Admin](#) — Preview.

Function — Commit

Persist rows. Returns import_id for rollback.

Per-role behavior:

[Admin](#) — Commit.

Function — Rollback

Reverse a prior import within the rollback window.

Per-role behavior:

[Admin](#) — Rollback.

12. Settings (per tile)

Settings hub launches every per-school configuration tile. Tiles a user can't open are hidden, so the visible set is their permission profile.

Settings Hub (launcher)

Nav path: Sidebar !' Settings

Audience: Admin / Core Team / SuperUser / District Admin

Purpose

Tile launcher grouped by Hall Pass Operations / School Identity & Schedule / Family & Signage / Feature Configuration / Admin & Tenancy.

Function — Open a tile

Routes to the tile's editor.

Per-role behavior:

Admin — See all tiles.

Core Team — See subset.

Bell Schedules

Nav path: Settings !' Bell Schedule

Audience: Admin / Behavior Specialist / MTSS Coordinator

Purpose

Define periods. Mark one schedule as default — required for the Hall Pass Queue's period-based reset.

Function — Create / edit schedule

Period list with start/end times.

Per-role behavior:

Admin / BS / MTSS — Edit.

Function — Mark default

Drives queue reset.

Per-role behavior:

Admin / BS / MTSS — Mark.

Locations & Allowlist

Nav path: Settings !' Locations / Allowlist / Teacher Allowlist

Audience: Admin

Purpose

Control valid hall pass destinations and which teachers may receive passes.

Function — Manage location list

CRUD on locations.

Per-role behavior:

Admin — Edit.

Function — Manage destination allowlist

Per-teacher receivable destinations.

Per-role behavior:

Admin — Edit.

School (identity)

Nav path: Settings !' School

Audience: Admin

Purpose

School name, address, default pass limit, time zone references.

Function — Edit fields

—

Per-role behavior:

[Admin](#) — Edit.

School Branding & Logo Generator

Nav path: Settings !' Branding / Logo Generator

Audience: Admin

Purpose

Upload school logo, set color palette, generate alternate logo variants.

Function — Upload logo

Object Storage backed; school-scoped ACL.

Per-role behavior:

[Admin](#) — Upload.

Function — Generate variants

AI-generated logo variants.

Per-role behavior:

[Admin](#) — Generate.

PBIS Thresholds

Nav path: Settings !' PBIS Thresholds

Audience: Admin / PBIS Coordinator

Purpose

Tune house standings cutoffs, store affordability thresholds, milestone bands.

Function — Edit thresholds

Persists to school_settings.

Per-role behavior:

[Admin / PBIS Coordinator](#) — Edit.

School Feature Configuration

Nav path: Settings !' School Features

Audience: Admin

Purpose

Master switches: Hall Passes / Tardy Pass / Family Comm / PBIS / School Store / Accommodations / Log Intervention / Request Pullout / MTSS Plans / Bell Schedule / Displays.

Function — Toggle feature

Off !' corresponding sidebar item hidden for everyone in the school. Used to onboard a school in stages.

Per-role behavior:

[Admin](#) — Toggle.

Parent Portal Sections

Nav path: Settings !' Parent Portal Sections

Audience: Admin

Purpose

Configure which HeartBEAT sections (PBIS, hall passes, tardies, accommodations, staff notes) parents see.

Function — Toggle section

Per-section on/off.

Per-role behavior:

[Admin](#) — Toggle.

School-wide Expectations

Nav path: Settings !' Expectations

Audience: Admin / PBIS Coordinator

Purpose

Define the school's PBIS expectations matrix (e.g. Be Respectful / Be Responsible / Be Safe across locations).

Function — Edit matrix

—

Per-role behavior:

[Admin / PBIS Coordinator](#) — Edit.

Intervention Strategies

Nav path: Settings !' Intervention Strategies

Audience: Core Team

Purpose

Strategy categories + strategies for the Tier 3 weekly form.

Function — CRUD on categories / strategies

—

Per-role behavior:

[Core Team](#) — Edit.

ISS Settings

Nav path: Settings !' ISS Settings

Audience: Admin

Purpose

Reasons catalog, per-day period count, roll-forward behavior.

Function — Edit

—

Per-role behavior:

[Admin](#) — Edit.

Staff Defaults

Nav path: Settings !' Staff Defaults

Audience: Admin

Purpose

Default permissions / capabilities applied to any newly-added staff member.

Function — Set defaults

—

Per-role behavior:

[Admin](#) — Edit.

Staff Directory

Nav path: Settings !' Staff Directory

Audience: Admin

Purpose

List of all staff for the school with role pills and last-login.

Function — Open Roles Matrix

Bulk role granting/revoking.

Per-role behavior:

[Admin](#) — Open.

Camera Registry

Nav path: Settings !' Cameras

Audience: Admin (capability-flagged)

Purpose

Catalog of building cameras for footage requests in cases.

Function — Add camera

Name, location, identifier.

Per-role behavior:

[Admin](#) — Add.

Case Outcomes Catalog

Nav path: Settings !' Case Outcomes

Audience: Admin / SuperUser / District Admin

Purpose

Per-school configurable catalog of case-closure outcomes. Closing a case requires picking from this catalog (no skip).

Function — List active outcomes

Default seeded set: no_action / conflict_resolution / mediation / parent_contact / office_referral / iss_assigned / oss_assigned / safety_plan_update / other.

Per-role behavior:

[Admin](#) / [SuperUser](#) / [District Admin](#) — View.

Function — Add / edit / disable

Inline edit label + description + sort order; toggle active. Disabling retires the outcome from new closures but preserves historical citations.

Per-role behavior:

[Admin](#) / [SuperUser](#) / [District Admin](#) — Edit.

Tenancy

Nav path: Settings !' Tenancy

Audience: SuperUser only

Purpose

Add/rename schools, manage districts.

Function — Add school

Per-role behavior:

SuperUser — Add.

Function — Rename / archive

Per-role behavior:

SuperUser — Edit.

Notifications

Nav path: Settings !' Notifications

Audience: Admin

Purpose

Pending admin notifications (alerts that require attention) + email-reminder configuration.

Function — Read / dismiss

Per-role behavior:

Admin — Manage.

Staff Preview ('view-as')

Nav path: Settings !' Staff Preview

Audience: Admin

Purpose

Render the app as if signed in as another staff role to verify what they see (no impersonation in writes).

Function — Pick a role profile

Read-only preview lens.

Per-role behavior:

Admin — Preview.

School Plans (safety plan library)

Nav path: Settings !' School Plans

Audience: Guidance Counselor + Core Team

Purpose

School-wide safety plan item library + categories.

Function — CRUD on items

Per-role behavior:

Guidance Counselor + Core Team — Edit.

13. Parent Portal

Standalone /parent/* application bundled with the staff app. Parents sign in via magic-link invites managed by Admins.

Parent Access (admin tooling)

Nav path: Sidebar !' Family Communication !' Parent Access

Audience: Admin

Purpose

Manage parent invites, link parents to students, configure which sections each parent sees, send invitation emails.

Function — Invite parent

Email + link to one or more students.

Per-role behavior:

Admin — Invite.

Function — Revoke

Disable a parent's access.

Per-role behavior:

Admin — Revoke.

Parent HeartBEAT (the parent's view)

Nav path: /parent/* (separate from staff app)

Audience: Parent

Purpose

Read-only HeartBEAT data for their student(s).

Function — View HeartBEAT

PBIS, hall passes, tardies, accommodations, staff notes — exactly the sections enabled in Settings !' Parent Portal Sections.

Per-role behavior:

Parent — Read for any linked student.

Function — Sibling switcher

Switch between children when one parent account is linked to multiple students.

Per-role behavior:

Parent — Switch.

Function — Export PDF

Download a snapshot of the report.

Per-role behavior:

Parent — Export.

15. Parent Pick-Up Module

End-of-day dismissal coordination. An append-only event log (`pickup_queue_events`) is the source of truth for queue state; every UI surface is a projection of it. The whole module is gated by `canManagePickup` (admin + Core Team + counselor + front-office + confidential secretary; teachers excluded).

Curb Keypad

Nav path: /pickup/curb (standalone kiosk page)

Audience: Front-office, dismissal monitors, admins

Purpose

Phone-first numeric keypad. Office staff type the 4-digit pickup number from the car hanger; the page returns the primary student plus every sibling that shares the same authorized parent. Tapping 'Add to line' writes an `added` event for each student and they appear on the classroom signage tile so the teacher releases them.

Function — Type pickup number

4-digit input (range 1001–9999; schema is TEXT to allow a future 5-digit expansion). Matches `student_pickup_authorizations` within the active school.

Per-role behavior:

Admin, Front Office, Core Team, Counselor — Look up and enqueue.

Teacher — Page is hidden — `canManagePickup` excludes teachers.

Function — Sibling roll-up

After a match, every student linked to the same `parentId` on the typed tag is shown. Bulk 'Add' enqueues all siblings in one event burst.

Per-role behavior:

All authorized pickup staff — See and add the whole sibling group at once.

Notes:

- If an authorization has a null `parentId` (guardian-only tag with no portal account) sibling roll-up is skipped — only the primary student is returned.

Function — Restricted-tag override

If an authorization is flagged `restrictedFrom` (court order / no-contact), the lookup returns a red banner and refuses the add. Admins can override with a "e5-char justification; this writes a `restricted_override` event tagged with the actor + reason.

Per-role behavior:

Admin — Override with justification.

Front Office, Core Team, Counselor — See the banner; the Override button is hidden.

Notes:

- A non-override touch is still recorded as a `restricted_attempt` event for the audit trail.

Function — 'In car' terminal step (new)

When `schoolSettings.pickupInCarStepEnabled = true`, each enqueued student gets an explicit 'In car' button on the curb page; tapping it writes an `in_car` event and clears the row from every queue display. When the toggle is off, `released_to_walk` is treated as terminal and the row auto-fades after `schoolSettings.pickupWalkedOutDisplaySeconds` (60–1800; default 300).

Per-role behavior:

Admin, Front Office, Core Team, Counselor — Tap 'In car' to terminate.

Notes:

- Toggle + display seconds live on the Pickup Settings page (Section 15 !' Pickup Settings).
- The `/api/pickup/queue` response always includes `inCarStepEnabled` + `walkedOutDisplaySeconds` so

every consumer renders the same UX.

Function — Live queue strip

Bottom of the page shows students currently 'in line' or 'walking out' with elapsed time. Student avatars (StudentPhoto) appear at 72px for face-match.

Per-role behavior:

[All authorized pickup staff](#) — View.

Screen notes:

- Auth: standalone /pickup/* pages require a staff session. Without one, the page redirects to the main login.
- Every action writes to pickup_queue_events (append-only); the queue endpoint reduces today's events into the visible state. There is no UPDATE/DELETE path.

Walker Gate

Nav path: /pickup/walkers

Audience: Walker-gate staff, admins

Purpose

Releases students flagged dismissalMode = 'walker'. Same keypad and sibling logic as the curb page, but the release button is gated by a bell-window banner — the page refuses the release if the configured 'Walker Release' bell period is not active.

Function — Bell-window gate

Reads the active bell schedule + period. Banner shows the window; release button is disabled outside it.

Per-role behavior:

[Admin, Front Office, Core Team, Counselor](#) — Release inside the window.

Notes:

- Admins can manually override the window (writes the override into the audit trail).

Function — Release walker

Writes walker_released — terminal event; the student does not need an 'in car' tap.

Per-role behavior:

[Admin, Front Office, Core Team, Counselor](#) — Release any walker on the active sibling group.

Classroom Signage Tile

Nav path: /pickup/teacher (or Display playlist tile)

Audience: Teachers (read), admins

Purpose

TV signage tile inside the classroom that lights up when a curb operator enqueues a student in the teacher's roster. Filtered by playlist-owner roster, so each classroom only sees its own kids.

Function — Send to line

Teacher (or signage operator) taps to confirm release; writes released_to_walk and starts the 'walking out' clock.

Per-role behavior:

[Teacher](#) — Send any student on their roster.

Notes:

- 10-second undo window writes release_undone.

Function — Teacher view scope

schoolSettings.pickupTeacherViewScope = 'own_roster' restricts visibility to roster students; 'all_students' shows the school-wide queue (small-school mode).

Per-role behavior:

Admin — Pick the scope in Pickup Settings.

Pickup Tags Admin

Nav path: Sidebar !' Settings !' Pickup Tags (or /pickup/admin)

Audience: Admin + canManagePickup

Purpose

Issue, reprint, and restrict the 4-digit car-tag numbers; manage extra-guardian splits; print physical PDF tags with QR codes.

Function — Bulk start-of-year assign

Walks the active roster and assigns the next free number per family (siblings share a number by default). Skips students who already hold an active tag.

Per-role behavior:

Admin — Run.

Function — Lost-tag reissue

Voids an existing number and assigns the next free one. The old number is held in a cooldown so it doesn't get re-handed-out immediately.

Per-role behavior:

Admin — Reissue with reason.

Function — Extra-guardian split

Issues a second authorization on a different number for a non-cohabiting guardian; both tags carry the same studentId.

Per-role behavior:

Admin — Issue.

Function — Print PDF — single + batch

Server renders 2×2 grid (4 tags/page) with student name, family label, QR code, and a prominent red RESTRICTED badge for restricted tags. Renderer lives in artifacts/api-server/src/lib/pickupTagsPdf.ts.

Per-role behavior:

Admin — Print individual tags or a full school batch.

Function — 80%-of-range capacity warning

NUMBER_RANGE_MAX = 9999. When used numbers cross 80% of the 8,999-slot range, the panel shows an amber warning to schedule the 5-digit expansion (future work in replit.md).

Per-role behavior:

Admin — See warning.

Function — Toggle restrictedFrom

Flips an authorization's restricted state. Curb-page lookups refuse non-admin pickup when this is true.

Per-role behavior:

Admin — Toggle.

Still-on-Campus Reconciliation Tile

Nav path: Admin Hub !' Still on Campus

Audience: Admin, front-office

Purpose

Post-cutoff reconciliation. After schoolSettings.pickupCutoffTime (default 15:30), the tile lists every student who has no terminal event today (in_car, walker_released, or auto_cleared), grouped by dismissalMode. Used to drive 'who do we call?' parent contacts.

Function — View grouped list

Groups by dismissalMode (car_rider, walker, bus, aftercare).

Per-role behavior:

Admin, Front Office — View.

Function — Mark auto-cleared

Admin can mark a remaining student as auto_cleared with a note (e.g., 'parent confirmed off-campus pickup'). Writes an auto_cleared event.

Per-role behavior:

Admin — Clear with note.

Screen notes:

- Tile hidden before the cutoff time so it doesn't pollute the morning Admin Hub.
- Terminal logic respects pickupInCarStepEnabled — when off, released_to_walk also counts as terminal so the tile doesn't false-positive.

Pickup Settings

Nav path: Settings ! Pickup

Audience: Admin

Purpose

Per-school configuration tile for the dismissal module.

Function — Cutoff time

HH:MM — controls when the Still-on-Campus tile appears.

Per-role behavior:

Admin — Edit.

Function — Teacher view scope

'own_roster' vs 'all_students'.

Per-role behavior:

Admin — Edit.

Function — 'In car' terminal step toggle (new)

Boolean. When true, every released student must be tapped 'in car' before the row clears. When false, released_to_walk is terminal and rows auto-fade after the configured display seconds.

Per-role behavior:

Admin — Toggle.

Function — Walked-out display seconds (new)

Integer 60–1800, default 300. Only takes effect when the in-car toggle is off. PUT validates the range; out-of-range payloads 422.

Per-role behavior:

Admin — Edit.

16. Student Photos & ID Badges

students.photo_object_key + students.photo_consent live on the canonical student row. Photos are stored via the school-scoped object-storage ACL (bindObjectToSchool) and rendered through the StudentPhoto component everywhere a student avatar appears. The badge generator embeds them into printable IDs.

Student Photo upload

Nav path: Student profile !' Photo tile (per-student); bulk via Data Importer 'yearbook zip'

Audience: Admin, Core Team (canManageStudentPhoto)

Purpose

Single-entry ingestion path today: per-student in-browser capture using getUserMedia (cropped to a square) or file picker. Result is uploaded through /api/storage/* and the returned object key is bound to the school's ACL. Bulk yearbook-ZIP ingest is documented as future work in the route header.

Function — Upload (per-student)

POST /api/students/:studentId/photo (replace) and DELETE /api/students/:studentId/photo (clear). Accepts an object key from /api/storage/upload, calls bindObjectToSchool, writes students.photo_object_key. Photo-consent toggle lives at PATCH /api/students/:studentId/photo-consent (admin-only).

Per-role behavior:

Admin, Core Team — Upload, replace, remove.

Teacher — Hidden.

Notes:

- Bind step enforces the school-scoped ACL; without it the asset is unreachable to the badge renderer and StudentPhoto.

Function — Camera capture

getUserMedia !' square crop !' blob upload !' bind. Falls back to file picker on browsers without camera access.

Per-role behavior:

Admin, Core Team — Capture in browser.

Function — Consent toggle

students.photo_consent (BOOL DEFAULT true). When false, every StudentPhoto surface renders initials regardless of whether a file exists. The file is not deleted — flipping the toggle back restores the photo.

Per-role behavior:

Admin — Toggle on student profile.

Screen notes:

- Photo-key column is nullable. The fallback is the existing initials bubble (consistent color per name).
- Parent portal never sees the photo — the staff-only ACL blocks it even if a parent guessed the URL.

Where the photo renders

Nav path: Cross-cutting

Audience: All staff

Purpose

StudentPhoto is the single component; every surface uses it. Replacing the avatar in one place automatically updates everywhere.

Function — Surfaces

Teacher roster avatars, PBIS Hub cards, Spotlight reveal, pickup curb confirmation (72px), walker gate, Watchlist, safety-plan picker, ID badges.

Per-role behavior:

All staff — See where the surface is otherwise visible.

Student ID Badges (PDF)

Nav path: Settings !' Student ID Badges

Audience: Admin

Purpose

Renders printable badges (lanyard portrait or CR80 credit-card landscape) with student photo (when consent), name + ID, house ribbon, and a Code128 + QR encoding the student_id for kiosk scans.

Function — Render PDF (single or batch)

GET /api/students/id-badges.pdf and POST /api/students/id-badges.pdf both call the same handler. Accepts a student-id list (or a scope: grade / homeroom / school) and a size (lanyard portrait or CR80 landscape). Bounded concurrency (6) and 4MB per-photo cap keep the renderer from OOMing on 'print all'.

Per-role behavior:

Admin — Print.

Function — Audit ledger

Every print writes a badge_print_events row: actor, students printed, batch size, reason (lost / damaged / first-issue / reprint). Read via GET /api/students/badge-print-events. Best-effort — a failed audit insert does not block the PDF.

Per-role behavior:

Admin — View audit table.

17. AST + Comp Time — Staff Time Banks

Two parallel time banks. AST (Alternate Schedule Time) is the HCTA-contract earn-then-use bank for exempt instructional staff. Comp Time (FLSA compensatory time, 1.5x earn rate, 240h cap) is the non-exempt bank — visible only to staff whose `exempt_status='non_exempt'`. Exempt staff hitting /comp see a 'not eligible — use AST instead' splash. Both banks store quarter-hour INTEGER units (no float drift); both are keyed to `staff_id` (not (`school_id`, `staff_id`)) so balances follow a staff member across schools within the same district. The Non-Exempt role preset collapses the sidebar to Hall Pass + Tardy Pass + Comp Time only, since those are the only three surfaces this role ever uses.

Staff AST Page

Nav path: Sidebar !' AST (AstSidebarBadge shows unread admin decisions)

Audience: All staff

Purpose

Personal ledger + request submission. Shows district-wide banked total, YTD earned/used, and full ledger history.

Function — Submit Earn request

Pre-approval workflow: staff submits intent (hours + category + description). Admin pre-approves, work happens, staff hits 'Submit completion' with actual hours, admin confirms — only the confirm step writes the ledger credit (`earn_confirm`).

Per-role behavior:

All staff — Submit.

Function — Submit Use request

Single-step: staff requests time off; admin approval writes the ledger debit (`use_approval`).

Per-role behavior:

All staff — Submit.

Function — Cancel pending

Staff can cancel an earn request before completion confirm and a use request before approval. After approval the debit is posted and only admin intervention can reverse it.

Per-role behavior:

All staff — Cancel within window.

Function — Acknowledge bell

On mount `/api/ast/acknowledge` clears the AstSidebarBadge.

Per-role behavior:

All staff — Auto.

Screen notes:

- Categories (Family-Facing / Athletics / Curriculum / etc.) are admin-set and stripped from staff-facing responses to avoid 'why was my work classified as X' disputes.

Admin AST Approval Queue

Nav path: Admin Hub !' AST Approval Queue (AstNotificationBell pulses when non-empty)

Audience: Admin / District Admin / SuperUser / staff with `canApproveAst` (confidential secretary)

Purpose

Two-stage approval queue for Earn (pre-approve + confirm) and one-step approval for Use.

Function — Earn pre-approval

Approve / deny before work happens. Must pick a category on approve. Denial requires a note (so the

staff member can re-submit).

Per-role behavior:

Approver — Pre-approve + categorize, or deny with note.

Function — Completion confirm

After staff submits actual hours, admin confirms or amends; confirm writes the earn_confirm ledger credit (the only credit path).

Per-role behavior:

Approver — Confirm; amends require a justification note.

Function — Use approval

One-tap approve or deny-with-note. Approve writes the use_approval debit.

Per-role behavior:

Approver — Approve/deny.

AST Insights

Nav path: Admin Hub !' AST Insights

Audience: Approver-tier

Purpose

School-wide liability and usage analytics so admins can size the bank.

Function — Headline tiles

Live banked total, earned YTD, used YTD.

Per-role behavior:

Approver — View.

Function — Top 5 leaderboards

Top balances and top earners.

Per-role behavior:

Approver — View.

Function — By category + by role

Earned/used breakdown by AST category and by role-group (Admin / Core Team / Teacher / etc.).

Per-role behavior:

Approver — View.

Function — Monthly trend

Earned vs used bar chart (12-month rolling).

Per-role behavior:

Approver — View.

Lapse Cron (July 1)

Nav path: artifacts/api-server/src/cron/astLapse.ts

Audience: Engineering

Purpose

Zeros every positive balance on July 1 @ 00:05 local. Writes a 'lapse' ledger row per staff with a negative delta equal to their prior balance. Guarded by a year-specific pg_advisory_xact_lock so duplicate triggers cannot double-lapse.

Function — Run

Annual; idempotent.

Per-role behavior:

System — Run.

Screen notes:

- Open follow-ups (replit.md): voluntary mid-year transfer hook (transfer_lapse kind reserved; not yet wired), optional weekly Friday digest gated on per-school ast_email_digest_enabled, per-staff ledger drilldown GET /api/ast/staff/:id/ledger.

18. Feature Licensing & Plans

Two-tier flag model: a SuperUser-controlled 'super feature' (from plan + per-school overrides) AND a school-level admin toggle. `loadEffectiveFeatures(schoolId)` is the single read path and is cached per-request. The 'AND fix' means turning a school's admin toggle ON cannot enable a feature the plan does not license.

Plans (global catalog)

Nav path: /admin/feature-licensing !' Plans

Audience: SuperUser (cross-district required for create/edit/delete)

Purpose

Global catalog of bundles (Bronze/Silver/Gold/Enterprise — labels are tenant-defined). Each plan toggles a set of `super_feature_*` booleans and carries an optional quotas JSONB.

Function — Create / edit / delete plan

JSONB blob: { features: {...bools}, quotas: { maxPlaylists: N, maxParentAccounts: N, ... } }

Per-role behavior:

Cross-district SuperUser (`ALLOW_CROSS_DISTRICT_SUPERUSER=1`) — Full CRUD.

District SuperUser — Read-only.

Function — Assign plan to school

Writes the plan id to `schools.feature_plan_id` and recomputes the school's `super_feature_*` boolean snapshot inside a locked transaction (`lockSchoolForLicensing !' SELECT FOR UPDATE` on schools).

Per-role behavior:

SuperUser — Assign.

Per-School Overrides

Nav path: /admin/feature-licensing !' Schools !' Overrides drawer

Audience: SuperUser

Purpose

Per-school exceptions to the plan, with optional expiration (e.g., 30-day trial of an upsell module). Each upsert/delete writes a `feature_licensing_audit_log` row.

Function — Add / edit / remove override

Force a `super_feature` on or off and optionally set `expires_at`.

Per-role behavior:

SuperUser — Edit.

Function — Expiration sweep cron

`featureLicensingOverrideSweep.ts` revokes expired overrides and writes an `override_expired_sweep` audit row. Sweep is idempotent via the audit log dedup.

Per-role behavior:

System — Run.

Function — Bulk overrides (Phase 5)

`BulkOverridesPanel` applies a single override to every school in a district or platform-wide. Cross-district guard requires `ALLOW_CROSS_DISTRICT_SUPERUSER`.

Per-role behavior:

Cross-district SuperUser — Bulk apply.

Quota Telemetry & Enforcement

Nav path: /admin/feature-licensing !' Quota Telemetry

Audience: SuperUser

Purpose

KNOWN_SEAT_QUOTAS (currently maxPlaylists, maxParentAccounts) are enforced at the consumer-route level via checkQuota(schoolId, feature, quotaName); a 403 with { error: 'quota_exceeded' } is returned when a school would exceed its plan.

Function — Telemetry table

Lists schools at "e 80% of any quota (adjustable threshold). Sourced from getQuotaUsage.

Per-role behavior:

SuperUser — View.

Screen notes:

- Open follow-up: wire a third quota consumer to keep KNOWN_SEAT_QUOTAS honest (good candidates per replit.md: mtss.maxActivePlans or displays.maxConcurrentSchedules).

Licensing Audit Log

Nav path: /admin/feature-licensing !' Audit Log

Audience: SuperUser

Purpose

Read-only history of every licensing event: plan_assigned, override_upserted, override_deleted, override_expired_sweep. Used for idempotency of the sweep and to answer 'why does school X have feature Y?' questions.

Function — Browse

Most-recent-first list with filters by school + event kind.

Per-role behavior:

SuperUser — View.

School Feature Configuration (admin-side)

Nav path: Settings !' School Features

Audience: Admin

Purpose

School-level toggles. The visible toggle for a feature is greyed out when super_feature is OFF (the plan does not license it) and the legend explains 'Contact your district to enable'.

Function — Toggle a feature

Writes school_settings.feature_*. Effective state is the AND of super_feature_* (plan + override) and feature_*. Sidebar items, Quick Access promotions, and dashboard tiles all consult loadEffectiveFeatures.

Per-role behavior:

Admin — Toggle features the plan licenses.

SuperUser — Toggle anything (admin tier is implicit).

Function — Bulk feature picker

One-click enable-all / disable-all (within the plan's licensed set). Useful for end-of-summer reset.

Per-role behavior:

Admin — Apply.

19. SuperUser & District Admin Tenancy

Cross-school operating surface. District-scoped SuperUsers (default) can see every school in their home district; cross-district SuperUsers (ALLOW_CROSS_DISTRICT_SUPERUSER=1) get the platform tier (global plans, bulk overrides). District Admins can read across the district but do not mutate licensing or tenancy.

SuperUser Home (Rollups)

Nav path: /admin/superuser

Audience: SuperUser

Purpose

Aggregate counts (districts, schools, students, staff) + per-district summary. Landing page after multi-school login.

Function — Switch into a school

Sets req.schoolId for subsequent requests; every read re-binds.

Per-role behavior:

SuperUser — Switch to any school the role can see.

Function — Open Audit & Health panel

7-day activity timeline across licensing changes, ISS log edits, case events. Surfaces 'is anything on fire?' at-a-glance.

Per-role behavior:

SuperUser — View.

District Overview

Nav path: /admin/district-overview

Audience: SuperUser + District Admin

Purpose

Per-school detail for the selected district with side-by-side 7-day stats (PBIS events, hall passes, ISS days, active cases, parent invites). Used for cross-school comparison without a context switch.

Function — Per-school rollup row

Click-through opens the school's Admin Hub in a context switch.

Per-role behavior:

SuperUser — Switch to any row.

District Admin — View only — switch is 403 from the API.

Onboard-a-District Wizard

Nav path: Tenancy Panel !' Onboard a district

Audience: Cross-district SuperUser

Purpose

Atomic creation of a district + primary school + initial admin staff + default settings + plan assignment, all inside a single DB transaction so a partial failure rolls back the entire tenant.

Function — Run wizard

Multi-step modal: district details !' primary school !' admin email !' plan pick !' confirm. Generates a CSPRNG temp password via generateAndHashTempPassword and surfaces it once to the SuperUser for hand-off.

Per-role behavior:

Cross-district SuperUser — Run.

Onboard-a-School (existing district)

Nav path: Tenancy Panel !' Onboard a school

Audience: SuperUser

Purpose

Adds a school under an existing district. Reuses the temp-password flow.

Function — Add school

Creates schools row, default school_settings row, initial admin staff, plan assignment.

Per-role behavior:

SuperUser — Add.

Edit + Soft-Delete Districts / Schools

Nav path: Tenancy Panel rows

Audience: SuperUser

Purpose

Rename, change district binding, or deactivate. Soft-delete = active = false; data stays for audit.

Function — Edit name / metadata

Inline.

Per-role behavior:

SuperUser — Edit.

Function — Soft-delete

Sets active = false. Every read path joins on (school.active AND district.active) so a retired tenant cannot serve orphaned sessions.

Per-role behavior:

SuperUser — Soft-delete.

Screen notes:

- Cross-silo guard: assertSchoolInCallerDistrict prevents a district-scoped SuperUser from acting on schools they shouldn't see, even by guessing IDs.

Data-Integrity Check (Tenancy Panel)

Nav path: Settings !' Tenancy

Audience: SuperUser

Purpose

Reports orphans (rows where school_id IS NULL) and a per-school row-count grid across 15 major tables (students, staff, hall_passes, pbis_entries, ...). Sanity check after a botched import.

Function — Run check

Read-only snapshot.

Per-role behavior:

SuperUser — Run.

Function — Add school (inline)

Quick form alternative to the wizard.

Per-role behavior:

SuperUser — Add.

Reset to Temp Password

Nav path: Staff & Roles !' row !' 'Reset to temp password'

Audience: Admin (own school), SuperUser (any school)

Purpose

Generates a fresh CSPRNG temp password for a staff member, hashes it, writes the hash + must_change_at_next_login flag, and returns the plaintext ONCE to the actor for hand-off.

Function — Reset

Confirmation modal explains the temp password will be shown only once. After dismiss it cannot be retrieved.

Per-role behavior:

Admin — Reset any staff in their school.

SuperUser — Reset anywhere.

Screen notes:

- Helper: artifacts/api-server/src/lib/tempPassword.ts !' generateAndHashTempPassword. Same helper is used by both onboarding wizards.

20. Additional Admin Catalogs & Tiles

Single-purpose Settings tiles. Most are CRUD on a per-school catalog; many are referenced by daily-ops screens.

School Branding

Nav path: Settings !' School Branding

Audience: Admin

Purpose

Per-school header gradient, logo, and color overrides applied to printouts, the HeartBEAT parent snapshot, and the Kiosk masthead.

Function — Edit colors

Hex inputs validated against `/^#[0-9a-fA-F]{6}$/`.

Per-role behavior:

Admin — Edit.

Function — Upload logo

POST `/api/school-branding/logo/bind` — binds the uploaded object key to the school ACL.

Per-role behavior:

Admin — Upload.

Screen notes:

- `gradient_colors` persists as a JSON string for portability across drizzle-kit versions.

Logo Generator

Nav path: Settings !' Logo Generator

Audience: Admin / SuperUser

Purpose

Pure client-side tool that produces Pulse-branded SVG / PNG assets for sister apps (PulseTV, Kinetics, Athletics). No server round-trip.

Function — Pick preset

Bundled color packs.

Per-role behavior:

Admin — Generate.

Function — Export SVG / PNG

Canvas rasterization at 4× for retina.

Per-role behavior:

Admin — Download.

Screen notes:

- SVG animation does not survive PNG export — a static fallback path is used in the raster.

Heartbeat Sections Admin

Nav path: Settings !' Heartbeat Sections

Audience: Admin

Purpose

Picks which 'Today's Heartbeat' signage tiles render and in what order (PBIS events, hall passes today, kiosk pulse, weather, etc.).

Function — Toggle / reorder

Drag-and-drop with per-section visibility.

Per-role behavior:

[Admin](#) — Edit.

School Closed Days

Nav path: Settings !' School Closed Days

Audience: Admin / Core Team

Purpose

Non-instructional day calendar. ISS day-rollover and Add-Discipline-Log calendars skip these dates so 'no school' days don't show up as missed attendance.

Function — Add / remove

Date + label.

Per-role behavior:

[Admin](#), [Core Team](#) — Edit.

Screen notes:

- Read is open to any signed-in staff.

Pullout Note Templates

Nav path: Settings !' Pullout Note Templates

Audience: Core Team

Purpose

Canned parent-message templates with placeholders the Verify modal substitutes client-side: {firstName} {lastName} {teacherName} {reason} {period} {schoolName}.

Function — CRUD

Mirrors pulloutReasons.ts pattern.

Per-role behavior:

[Admin](#), [Behavior Specialist](#), [MTSS Coordinator](#), [Dean](#) — Edit.

Trusted Adult Links

Nav path: Settings !' Trusted Adults

Audience: Core Team

Purpose

Manual link between staff and student that grants the staff Watchlist + Insights visibility regardless of the section roster. Used for advisor / mentor relationships.

Function — Add / remove

Inline.

Per-role behavior:

[Admin](#), [Core Team](#), [PBIS Coordinator](#) — Edit.

Trusted-Adult Interventions

Nav path: Settings !' Trusted-Adult Interventions

Audience: Core Team

Purpose

Catalog of intervention types that trusted-adult mentors can log (separate from teacher tier 2/3).

Function — CRUD

—

Per-role behavior:

[Admin, Core Team](#) — Edit.

Separation Tags Admin

Nav path: Settings !' Separation Tags

Audience: Admin

Purpose

Tag catalog used by SuggestSeparationModal and the Separation Suggestions aggregate.

Function — CRUD

—

Per-role behavior:

[Admin](#) — Edit.

Polarity Pairs

Nav path: Sidebar !' Interventions !' Polarity Pairs

Audience: Core Team

Purpose

Two students who must NOT both be out on a hall pass at the same time. findPolarityConflict is called by hallPasses + kiosk pass creation — the rule is enforced at issuance, not just by convention.

Function — CRUD pair

Pick both students + reason.

Per-role behavior:

[Admin, Behavior Specialist, Dean, MTSS Coordinator](#) — Edit.

Screen notes:

- The conflict short-circuits new-pass creation with a 409 + the other student's name.

Discipline Reasons Catalog

Nav path: Settings !' Discipline Reasons

Audience: Admin

Purpose

Dropdown source for ISS / OSS logging.

Function — CRUD + active toggle

Inactivating preserves historical labels.

Per-role behavior:

[Admin](#) — Edit.

Custom Roles

Nav path: Settings !' Custom Roles

Audience: SuperUser

Purpose

District-defined named role profiles built from capability flags (cap_*). Lets a district mint, e.g., 'Attendance Clerk' without code changes.

Function — Create role profile

Name + capability set.

Per-role behavior:

SuperUser — Create.

Function — Assign to staff

From Staff & Roles.

Per-role behavior:

Admin — Assign existing profiles.

FAST Coverage

Nav path: Settings !' FAST Coverage

Audience: Admin

Purpose

Pre-flight check before staff use the Roster: per-grade / per-subject count of students with PM1/PM2/PM3 scores vs total roster.

Function — View coverage

Status badges flag 'Missing PM3' / 'Partial PM3' in red/amber.

Per-role behavior:

Admin — View.

Screen notes:

- A 'No chart' flag means the subject is imported but FL DOE cut-scores aren't wired yet — common for Geometry until the next import cycle.

Kiosk Setup & Activation

Nav path: Settings !' Kiosk Setup

Audience: Admin / Core Team

Purpose

QR/PIN activation cards for classroom tablets. Each card is a kiosk_enroll_token; exchange writes a kiosk_activation + a kiosk_viewer_token (long-lived but rotatable).

Function — Bulk generate teacher cards

One-click PDF for every active teacher.

Per-role behavior:

Admin — Generate.

Function — Activate proxy (sub coverage)

Core Team can pre-activate a kiosk for a substitute for today or 14 days.

Per-role behavior:

Core Team — Activate.

Function — Token rotation on print

Printing a card invalidates the previous token immediately — security feature; hand out new cards right after printing.

Per-role behavior:

Admin — Print.

Camera Registry + Scanner

Nav path: Settings !' Camera Registry

Audience: Admin / Dean

Purpose

Catalog of school security cameras (id, label, location, retention days, viewer URL). Used by the Investigations Case Detail to issue footage requests.

Function — CRUD camera

—

Per-role behavior:

Admin, Dean — Edit.

Function — Scan QR (CameraScanner)

Mobile QR-scan helper to onboard a camera by sticker.

Per-role behavior:

Admin, Dean — Use.

Email Digest & Preview

Nav path: Settings !' Notifications !' Email Digest

Audience: Admin

Purpose

Preview today's digest email (rendered HTML) and force-fire the dispatcher for the school. Backed by /api/email-preview and /api/digest.

Function — Preview

Renders the current digest with live data.

Per-role behavior:

Admin — Preview.

Function — Send now

Manual trigger — not rate-limited; use sparingly.

Per-role behavior:

Admin — Send.

Screen notes:

- Cron is gated on EMAIL_REMINDERS_ENABLED + RESEND_FROM_ADDRESS (see replit.md Gotchas).

School Store Editor

Nav path: Sidebar !' School Store

Audience: Admin / PBIS Coordinator (edit); all staff (read)

Purpose

School-wide reward catalog (separate from the per-teacher Classroom Store). Items support image, cost (PBIS points), inventory, and active toggle. Teachers see the same catalog read-only.

Function — CRUD item

Image uploads use /api/storage/* with school-scoped ACL bind (bindObjectToSchool). pendingUploads ledger orphans cleanup.

Per-role behavior:

Admin, PBIS Coordinator — CRUD.

Teacher — Read-only.

Function — Toggle active

Inactive items disappear from the redemption picker but historical redemptions keep their label.

Per-role behavior:

Admin, PBIS Coordinator — Toggle.

21. Insights — Additional Dashboards & Drilldowns

Surfaces that complete the Insights suite already documented in Section 7. All Insights endpoints are core-team gated and accept grades + window filters; CSV export is described in Section 22.

SEB/SEL Dashboard

Nav path: Sidebar !' Insights !' SEB/SEL

Audience: Core Team

Purpose

Whole-school social-emotional / behavioral lens. Pulls active student_mtss_plans (bucketed into 5 plan-area categories), ESE / 504 / ELL flags, last-30-day negative PBIS, FAST priorYearBq, and active accommodations.

Function — Filter grades

Defensive parser; bad input becomes 'no filter'.

Per-role behavior:

Core Team — Filter.

Function — Export CSV

See Section 22.

Per-role behavior:

Core Team — Export.

Screen notes:

- Time window is fixed at 30 days for the negative-PBIS signal; every other signal is stateful.

Attendance Dashboard

Nav path: Sidebar !' Insights !' Attendance

Audience: Core Team

Purpose

Absences + tardies per student over the chosen window with rate calculation and top-N lists.

Function — Window picker

windowKey (week / month / quarter / custom).

Per-role behavior:

Core Team — Pick.

Function — Export CSV

See Section 22.

Per-role behavior:

Core Team — Export.

Academics Trajectory (longitudinal drill-in)

Nav path: Insights !' Academics !' Trajectory

Audience: Core Team

Purpose

PM1 !' PM2 !' PM3 trajectory per student per subject. Classifies students into archetypes (steady-high, declining, recovering, stuck-low) and sub-archetypes. Powers the BandStudentsDrawer drill-in.

Function — Pick subjects

Multi-select with band coloring.

Per-role behavior:

Core Team — Pick.

Function — Open band drawer (BandStudentsDrawer)

Click a chart band to see the constituent students.

Per-role behavior:

Core Team — Drill in.

Function — Export full CSV

Server-streamed; see Section 22.

Per-role behavior:

Core Team — Export.

Function — Export drawer CSV

Client-rendered limited to drilldown.

Per-role behavior:

Core Team — Export.

Insights Watchlist

Nav path: Insights !' Watchlist

Audience: Core Team

Purpose

Personal pinned-students view: each Core Team member can pin students they want to track; the dashboard aggregates each pinned student's PBIS, passes, ISS, MTSS, accommodations into one row.

Function — Pin / unpin

Toggles in MyWatchList.

Per-role behavior:

Core Team — Pin.

Separation Suggestions

Nav path: Insights !' Behavior !' Separation Suggestions

Audience: Scheduling team (admins, counselors, Core Team — read); Teachers (write only)

Purpose

Aggregates SuggestSeparationModal flags filed by teachers. Pairs with high teacher-consensus counts surface to the master-schedule builder for next year.

Function — Filter grade / consensus threshold

Min #teachers + grade.

Per-role behavior:

Admin, Counselor, Core Team — View.

Function — Drill into student

Full timeline of separation flags + notes.

Per-role behavior:

Admin, Counselor, Core Team — Drill.

Screen notes:

- Intended for next-year scheduling, not active behavioral intervention.

PBIS Goals + Milestones + Needs-Attention

Nav path: Sidebar !' PBIS Hub

Audience: PBIS Coordinator / Admin

Purpose

pbis_goals = per-student period goals (week/month/quarter/all). pbis_milestones = thresholds that trigger automated parent emails when crossed. PbisNeedsAttention surfaces 'quiet teachers' (no points awarded in X days), 'invisible students' (zero points in X days), and reason imbalance (>80% one reason).

Function — Create / archive goal

Creator or PBIS Coord/Admin can archive.

Per-role behavior:

Teacher, PBIS Coordinator, Admin — Create on their own students; PBIS Coord/Admin archive anyone's.

Function — CRUD milestone

Email template + threshold.

Per-role behavior:

Admin, PBIS Coordinator — Edit.

Function — Tune thresholds

Quiet-teacher days, invisible-student days, reason-imbalance ratio — Settings !' PBIS Thresholds.

Per-role behavior:

Admin, PBIS Coordinator — Edit.

Houses Panel + Change-House Modal + Sort Jobs

Nav path: Sidebar !' PBIS Hub !' Houses

Audience: PBIS Coordinator / Admin

Purpose

Manages the 4-house system. ChangeHouseModal records the move into student_house_changes (audit). Sort jobs (student_house_sort_jobs) batch-balance new students into houses.

Function — View standings

Live points + ranking (consumed by Spotlight governor v2 — see Section 3 / Spotlight).

Per-role behavior:

Any staff — View.

Function — Change house

Writes audit row with actor + reason.

Per-role behavior:

Admin, PBIS Coordinator — Change.

Function — Run sort job

Bulk-assign new students.

Per-role behavior:

Admin, PBIS Coordinator — Run.

MTSS Reports

Nav path: Sidebar !' Interventions !' Reports

Audience: MTSS Coordinator / Admin

Purpose

Quantitative health of the MTSS program: completion stats, weekly trend lines, Mon-Fri heatmap.

Function — Per-teacher completion

Logged vs expected entries, by subject.

Per-role behavior:

[MTSS Coordinator, Admin](#) — View.

Function — Trend line

Weekly T2 completion %, T3 average score.

Per-role behavior:

[MTSS Coordinator, Admin](#) — View.

Function — Heatmap

Day-of-week completion.

Per-role behavior:

[MTSS Coordinator, Admin](#) — View.

Screen notes:

- Tier 3 completion is binary per week; Tier 2 is ratio-based.

22. CSV Exports — Cross-cutting Reference

Every CSV download in PulseEDU. Two implementation styles: server-streamed (Content-Disposition: attachment, UTF-8 BOM for Excel) and client-rendered (downloadCsv helper in InsightsPicker.tsx). Server route + filename pattern + columns are listed per surface so QA can verify any export end-to-end.

Academics Trajectory — full export

Nav path: Insights !' Academics !' Trajectory !' 'CSV' button

Audience: Core Team

Purpose

Server-streamed full-cohort export. GET /api/insights/academics/trajectory/export.csv. Filename trajectory_<subjects>_<grade>_<YYYY-MM-DD>.csv. Columns: student_id, student_name, grade, subject, pm1, pm2, pm3, pm1_band, pm3_band, archetype, sub_archetype.

Function — Apply filters then export

Honors subjects, grades, ell, ese, is504.

Per-role behavior:

Core Team — Export.

Trajectory Band Drawer export

Nav path: Trajectory !' click chart band !' '+' CSV'

Audience: Core Team

Purpose

Client-rendered (downloadCsv) limited to the drilled archetype + subKey. Capped at CAP=200 students.

Function — Export

trajectory_<subjects>_<arch>_<date>.csv

Per-role behavior:

Core Team — Export.

Data Export Panel

Nav path: Settings !' Data Management !' Export

Audience: Importer roles (Admin)

Purpose

Generic exporter mirror of the importer. GET /api/data-imports/export?kind=<kind>&... Filename pulseedu-<kind>-<date>.csv (BOM prefixed). Kinds: rosters, behavior, fast_scores, fast_prior_year, assessments. Required columns are always re-injected even if the user deselects them.

Function — Pick kind + filters + columns

Filters per kind: grade, date, subject, noteType, assessmentName. Scope: school (default) or district (assessments only).

Per-role behavior:

Admin — Export.

Screen notes:

- Server uses Papa.unparse with quoting; prepends \uFEFF so Excel reads UTF-8 names correctly.

Skipped roster rows (per import job)

Nav path: Data Imports !' Job !' 'Download skipped rows'

Audience: Admin

Purpose

GET /api/data-imports/jobs/:id/skipped-houses.csv — re-emits the rejected rows with their original columns. Filename skipped-houses_<orig_filename>_job<id>.csv.

Function — Download

Job-scoped; no extra filters.

Per-role behavior:

Admin — Download.

Insights dashboards — CSV button

Nav path: Each dashboard's filter bar

Audience: Core Team

Purpose

Client-rendered (downloadCsv) using topListsToCsv concatenated-list format (one file, multiple sub-tables separated by blank line, with a 'list' discriminator column). Filenames <dashboard>_<grades>_<YYYY-MM-DD>.csv.

Function — Per-dashboard export

Attendance, Behavior, Engagement, Equity, SEB/SEL, Academics, Early Warning each have their own button. Honors the current grades + window filters.

Per-role behavior:

Core Team — Export.

Screen notes:

- Client triggers use authFetch (Bearer) rather than a bare <a href> so the iframe sandbox does not strip the auth cookie.
- RFC 4180 escape: any cell containing comma, quote, or newline is wrapped in quotes; literal quotes doubled.

23. Witness Statements & AI Consistency Check

Two recently shipped layers on top of the Investigations suite (Section 6). Data layer is live; UI surfacing has open follow-ups noted in replit.md.

Witness Statement Numbering

Nav path: Investigations !' Case Detail !' Statements list

Audience: Investigators (Admin / Dean / BS)

Purpose

Deterministic per-case statement IDs in the form CASE-YY-NNNN-WS-XX. Assignment is transaction-locked so concurrent statement creation cannot collide on a number.

Function — Assign number on attach

When a statement is attached to a Case (a row moves from `interaction_witness_statements` with `case_id` IS NULL to NOT NULL) the helper assigns the next XX inside a `SELECT...FOR UPDATE` on the case row. Order is `created_at` ASC.

Per-role behavior:

System — Assign.

Screen notes:

- Backfill task at deploy: walk existing attached statements per case ordered by `created_at` ASC. Open follow-ups: surface the formatted ID in PlayerDrawer header, Case Detail statements list, witness statement PDF/print, audit-log payload (copy-on-click).

AI Consistency Check

Nav path: Investigations !' Case Detail !' Consistency panel (investigator-gated)

Audience: Case investigators (Admin / SuperUser / District Admin / Dean / Behavior Specialist / MTSS Coordinator) via `adminGate !' isCaseInvestigator`

Purpose

Background AI evaluation that compares statements + video evidence + interaction descriptions on a case and flags contradictions. Runs are scheduled when a statement is completed or video evidence is added; findings + dismissals + audit are stored in `case_consistency_runs`, `case_consistency_findings`, `case_consistency_state`.

Function — Run manually

'Run check' button. Gated by `adminGate` (admin tier + Dean + BS + MTSS).

Per-role behavior:

Admin, Dean, Behavior Specialist, MTSS Coordinator — Trigger.

Function — Dismiss finding with reason

Required justification "e5 chars; writes to `case_consistency_state`.

Per-role behavior:

Admin, Dean, Behavior Specialist, MTSS Coordinator — Dismiss.

Screen notes:

- Open follow-ups (replit.md): (1) onboarding step 'Review Consistency Check guardrails' in Behavior & PBIS phase with an 'I understand' marker, Core Team audience; (2) Settings tile 'Consistency Check — this month' backed by `GET /api/watchlist/consistency-telemetry` (admin-gated; cheap COUNT/SUM grouped by current month).

ISS Admin Log Detail Drawer (edit / delete + audit)

Nav path: Admin Hub !' ISS log !' row !' Detail Drawer

Audience: Admin

Purpose

Edit / trim / delete a posted ISS admin log row with a required reason; the prior payload is written to `iss_admin_log_audit` so the change history is reconstructable.

Function — Edit row

Required reason "e5 chars.

Per-role behavior:

Admin — Edit.

Function — Delete row

Soft-delete; visible in audit.

Per-role behavior:

Admin — Delete.

Function — View audit history

Chronological before/after JSON.

Per-role behavior:

Admin — View.

Screen notes:

- Synthesized rows from admin-hub-logged days (`iss_attendance_day`) have negative IDs and are NOT editable from the ISS dashboard; they must be edited from Admin Hub.

24. Cross-cutting Concerns

Behaviors that are not a screen on their own but are visible across many screens. Worth a separate read so QA covers them on every relevant page.

Multi-tenancy & School Scoping

Nav path: Every server route, every dashboard query

Audience: All staff

Purpose

Every read/write filters by req.schoolId. SuperUser may operate across schools but the active context is still bound; District Admin works district-wide.

Function — Cross-school reads (District Admin)

District Overview aggregates every school side-by-side; underlying queries filter by district_id and group by school_id.

Per-role behavior:

District Admin — Read across district.

Screen notes:

- QA: every list view should be re-tested with a multi-school account context-switched mid-session to confirm the previous school's rows do not leak in.

Audit Trails

Nav path: Investigations Case Detail, ISS log (planned), interaction edits

Audience: Admin (read), all editors (write)

Purpose

Sensitive mutations write an audit row with actor, before/after JSON, and a required reason on certain actions (case reopen, future ISS edits).

Function — Reopen case

Required reason "e5 chars; audit payload includes prior outcome code.

Per-role behavior:

Admin, SuperUser, District Admin — Reopen + reason.

Function — Edit / dismiss with reason

Several flows already require a justification (statement dismiss, AI consistency dismiss). Planned: ISS edit/trim/delete with required reason.

Per-role behavior:

Admin — Edit + reason.

Feature Flags (per-school)

Nav path: Settings !' School Features

Audience: Admin

Purpose

Disabling a feature hides every entry point: sidebar item, Quick Access promotion, dashboard tile.

Function — Disable a feature

Server keeps data but UI is hidden.

Per-role behavior:

Admin — Disable.

How-to-use Help (HowToUseHelp)

Nav path: Inline on every major screen

Audience: All staff

Purpose

Per-screen collapsible 'How to use' panel with role-aware sections (RoleSection). Source of truth for in-app docs; should be refreshed any time a screen gains a new affordance.

Function — Read help

Click to expand.

Per-role behavior:

All staff — Read.

Maintenance

When you ship a new screen or function, append a row to this document — do not let it drift. The 'How to use' panels inside the app (HowToUseHelp / RoleSection) should mirror the per-role behavior recorded here.

Open follow-ups tracked in replit.md !' Future work: AI Consistency Check onboarding step + 'Consistency Check — this month' Settings tile; per-school IANA timezone column (replace DEFAULT_SCHOOL_TZ with a per-school column threaded through schoolYearLabelFor, seed case backfill, AST insights, lapse cron); refresh Core Team 'How this works' copy after the Phase 4 case enhancements (tagging, video evidence panel, AI consistency check, Case Insights dashboard) ship as a single pass; Pickup module 5-digit expansion (bump NUMBER_RANGE_MAX once a tenant exceeds ~7200 active tags + narrow the PDF font + accept 4-or-5-digit input on the curb keypad); curb-line audible chime (design open — leaning visual-only since high-volume schools would overlap); Student Photos rollout (the StudentPhoto component, photo_object_key + photo_consent columns, and per-student upload route are shipped, but bulk yearbook-ZIP ingestion and several delivery surfaces remain open per replit.md); Witness statement formatted-ID surfacing in PlayerDrawer header + Case Detail list + PDF + audit log payload, plus one-time backfill at deploy; AST voluntary-transfer zero-out hook (transfer_lapse kind reserved) + optional Friday digest gated on ast_email_digest_enabled + per-staff ledger drilldown GET /api/ast/staff/:id/ledger; Feature Licensing Phase 4 (wire a third quota consumer to keep KNOWN_SEAT_QUOTAS honest — mtss.maxActivePlans or displays.maxConcurrentSchedules — and per-feature usage sparklines in the SuperUser admin page).